

LimitIQ v2

Multi-source credit-line governance benchmark

Executive evidence

One pooled model evaluates harmonized adverse-credit outcomes across six independent source cohorts. Source labels have different events and horizons, so the score is not a common-horizon regulatory probability of default.

Evidence class	Verified result
Training union	1,869,548 rows; six independent cohorts
Champion	Histogram gradient boosting
Macro untouched test	ROC-AUC 0.6845; Brier 0.1390
Pooled untouched test	ROC-AUC 0.6699; Brier 0.1406
Synthetic scenario	252 increases; INR 6,411,977 contribution

Synthetic portfolio decisioning

Action	Accounts
Manual review	619
Increase 10%	18
No change	283
Freeze automatic increases	46
Increase 30%	216
Increase 20%	18

Current versus proposed exposure: INR 567,613,000 to INR 608,791,300. Simulated expected-loss proxy: INR 75,984,097 to INR 77,625,095. Simulated risk-adjusted return: 20.74%. These are deterministic assumption-driven outputs, not causal or realized impact.

Untouched test by source cohort

Source cohort	N	ROC	Brier
FICO Explainable ML / HELOC (geography undisclosed)	1,974	0.766	0.199
Give Me Some Credit (geography undisclosed)	30,000	0.853	0.050
Home Credit Default Risk (geography undisclosed)	61,502	0.517	0.074
Lending Club Loans 2007-2018 (US)	274,234	0.602	0.165
South German Credit Data	200	0.613	0.203
Default of Credit Card Clients (Taiwan)	6,000	0.757	0.143

Governance boundaries

The seeded random split measures within-source interpolation only. Lending Club dominates pooled rows, while macro metrics weight cohorts equally. Region is one-hot encoded; missingness can identify a source. The model does not establish unseen-country, out-of-time, Indian, IFRS 9, fair-lending, or production generalization.

Publication gate: blocked pending manual review of upstream or competition redistribution and derived-artifact terms for Give Me Some Credit, FICO HELOC, Lending Club, and Home Credit. The live v1 Taiwan demonstration remains separate until that gate is resolved.

LimitIQ is an educational portfolio demonstration using public and synthetic data. It is not a production credit-decision system and must not be used to make real lending decisions.